Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering

LAB MANUAL

Course No.	**:**	**CSE 4110**
Course Name	**:**	**Information System Analysis and Design Sessional**
Semester	**:**	**4^{th} Year Odd Semester**
Prepared By	**:**	**Emrana Kabir Hashi, Assistant Professor, Department of Computer Science & Engineering (CSE), Rajshahi University of Engineering & Technology (RUET).**

Information System Analysis and Design Laboratory Guidelines

A detailed and systematically organized version of the **Information System Analysis and Design (ISAD) Laboratory Guidelines** is presented below:

1. Before Coming to Lab

Preparation

- Read the **ISAD lab manual** and the **experiment sheet** carefully before attending the class.
- Understand the **objectives**, **tools**, and **expected outcomes** for the day's experiment.

Pre-lab Work

- Complete any **pre-lab assignments** or **system analysis tasks** assigned.
- Bring your **lab notebook** dedicated to ISAD experiments.

Conceptual Understanding

- Review the theoretical concepts related to the lab, such as:
    - System Development Life Cycle (SDLC)
    - Feasibility Study
    - Requirement Analysis
    - Data Flow Diagrams (DFDs)
    - Entity-Relationship (ER) Models
    - System Design and Prototyping

Technical Setup

- Ensure your computer environment is ready with necessary software tools such as:
    - Draw.io / Lucidchart / StarUML / Visual Paradigm
    - MS Excel / Google Sheets for evaluation matrices
    - Any prototyping or diagramming tools required for the lab tasks.

2. During Lab Session

A. Conduct

- Be **punctual**; attendance will be taken at the beginning.
- Maintain **discipline** and a **professional attitude** during lab hours.
- Avoid unrelated activities, browsing, or using mobile devices.
- Handle all lab computers and materials with care.
- **Do not install or modify software** without the instructor's permission.

B. Work Process

- Work **individually** or **in groups** as instructed for team-based system analysis or design tasks.
- Follow the **step-by-step procedure** in the lab manual and instructor's guidance.
- Apply appropriate **problem-solving and analytical techniques** to model real-world systems.
- Before asking for help, review your work, check logical and design consistency, and ensure that diagrams or documentation follow proper conventions.

- Record all **steps**, **diagrams**, **requirements**, **analysis results**, and **designs** in your lab notebook for report preparation.

3. After Completing Experiment

- **Demonstrate** your completed analysis, model, or prototype to the instructor for evaluation.
- Be ready to **explain your design**, **data flow**, **functional requirements**, and **justifications** for your design choices.
- **Submit** your lab report or documentation in the prescribed format.
- Ensure that all files (diagrams, reports, etc.) are properly **named and stored**.
- **Clean up** your workspace and **log off** from the system before leaving.

4. Lab Report Writing

Each lab report should be organized clearly and include the following sections:

1. **Experiment Name & Objective** – Title and aim of the experiment.
2. **Theory** – Concise explanation of key ISAD concepts (e.g., DFD, ER diagram, feasibility study).
3. **Procedure Followed** – Steps performed during the analysis and design process.
4. **Tools / Methods Used** – Software and techniques applied (e.g., Lucidchart, decision tables).
5. **System Models / Diagrams** – Include DFDs, ERDs, use case diagrams, and structure charts as applicable.
6. **Observations / Outputs** – Final design models, feasibility matrices, or prototypes.
7. **Conclusion** – Summary of outcomes and insights gained from the experiment.

5. Evaluation Criteria

Students will be assessed based on the following components:

Criteria	Description
Preparation	Pre-lab readiness, understanding of concepts, and pre-lab assignments.
Implementation / Execution	Ability to perform system analysis and design tasks effectively.
Understanding / Viva	Conceptual understanding and ability to explain work done.
Reporting	Clarity, accuracy, and completeness of documentation and diagrams.
Discipline & Teamwork	Punctuality, collaboration, and professional conduct in lab.

Note to Students:

The ISAD laboratory is designed to help you bridge theory and practice. Each session contributes to developing your analytical, design, and teamwork skills essential for real-world information system development.

Lab Curriculum

Table-1: CO-PO mapping for CSE 4110

CO	CO Statement	PO	Delivery Method	Assessments
CO1	Analyze information systems by applying problem-solving techniques.	PO2	☒ Discussion ☒ Interaction ☒ Lab Manual	☒ Quiz ☒ Viva ☒ Lab Report ☒ Presentation
CO2	Design and develop an information system	PO3	☒ Discussion ☒ Interaction ☒ Lab Manual	☒ Viva ☒ Lab Report ☒ Presentation
CO3	Solve real life complex problems individually and collaboratively in team settings.	PO9	☒ Discussion ☒ Interaction ☒ Lab Manual	☒ Viva ☒ Lab Report ☒ Presentation

Table-2: PO statements

PO1	**Engineering knowledge:** Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization respectively to the solution of complex engineering problems.
PO2	**Problem analysis:** Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO3	**Design/development of solutions:** Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PO4	**Investigation:** Conduct investigations of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO5	**Modern tool usage:** Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations.
PO6	**The engineer and society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems.
PO7	**Environment and sustainability:** Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts.
PO8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
PO9	**Individual work and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.

PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance:** Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

List of Experiments

Lab 1

Experiment [CO1:PO2]

Name of the Lab Experiment:

Identification of the Problems of a Targeted Information System

1. Introduction/Description

Information systems are designed to collect, process, store, and distribute information for decision-making and operational control in an organization. Before developing or improving such systems, it is essential to identify the existing problems, inefficiencies, or gaps within the targeted information system.

In this experiment, students will analyze a selected information system — preferably from a software firm, IT company, or any organization using a computerized system — to identify system elements, characteristics, and types. They will then map these findings onto the **System Development Life Cycle (SDLC)** phases to understand how such systems evolve from analysis to implementation and maintenance.

This experiment develops analytical and problem-solving abilities (CO1) by encouraging students to investigate real-world systems, identify bottlenecks, and suggest potential improvements based on structured system analysis techniques.

2. Requirements

Knowledge Requirements:

- Basic understanding of **information system components** (hardware, software, people, data, process).
- Familiarity with **System Development Life Cycle (SDLC)** models (Waterfall, Iterative, Agile, etc.).
- Understanding of **types of information systems** (Transaction Processing System, MIS, DSS, etc.).

Materials / Tools Required:

- Access to a real or simulated organizational information system.
- Internet and documentation tools (Google Docs, Word, Excel, or any CASE tool).
- Presentation and diagram tools (Lucidchart, Draw.io, or MS Visio).

3. Procedure

1. **System Selection**
    - Choose a **target organization** or system (e.g., software firm, hospital management system, library management system, or online booking system).
    - Collect background information such as system purpose, users, and scope.
2. **Identify System Elements**
    - Define the five main components:
        - **Hardware:** Devices and network infrastructure.
        - **Software:** Applications and tools used.
        - **People:** End-users, administrators, and IT staff.
        - **Data:** Information inputs, storage, and outputs.
        - **Processes:** Procedures and workflows followed.
3. **Classify the Information System**
    - Determine the **type** (e.g., Transaction Processing System, Management Information System, or Decision Support System).
    - Identify system **characteristics** such as automation, interactivity, and data flow.
4. **Analyze System Problems**
    - Identify current limitations (e.g., slow data retrieval, lack of integration, security flaws).
    - Use **problem-solving techniques** like cause-effect analysis or process mapping.
5. **Apply the SDLC Framework**
    - Map the observed issues to the appropriate **SDLC phase**:
        - **Requirement Analysis:** What does the system fail to deliver?
        - **Design:** Are workflows inefficient or poorly structured?
        - **Implementation:** Are there technical constraints or poor code practices?
        - **Testing:** Are errors or failures recurring?
        - **Maintenance:** Is support or updating inadequate?
6. **Document and Report**
    - Prepare a report summarizing identified issues, analysis techniques used, and possible improvement suggestions.
    - Include diagrams such as context diagrams or data flow diagrams to visualize findings.

4. Precautions

- Select a **realistic and understandable system** to ensure meaningful analysis.
- Ensure **authentic data collection** — avoid assumptions without evidence.
- Respect organizational confidentiality and **do not disclose sensitive information**.
- Maintain clarity when defining system boundaries and functionalities.

5. Conclusion/Discussion

This experiment helps students understand how to analyze and identify problems in existing information systems using systematic approaches. By applying the **SDLC framework**, they can relate theoretical concepts to practical system evaluation.

Through this process, students develop the ability to:

- Break down complex systems into manageable components.
- Identify key issues affecting system performance or user satisfaction.
- Suggest informed improvements that enhance efficiency and reliability.

Thus, this lab directly contributes to **CO1 (Analyze information systems by applying problem-solving techniques)** and aligns with **PO2 (Cognitive – Analyzing, Applying; Affective – Organize)** by integrating analytical thinking, structured documentation, and system understanding.

Lab 2

Experiment [CO1:PO2]

Name of the Lab Experiment:

Perform Initial Investigation of the Information System

1. Introduction / Description

The initial investigation phase is the foundation of system analysis and design. It involves a preliminary study of the existing system to understand its structure, functionality, limitations, and potential for improvement. The purpose of this lab is to help students identify how information flows within an organization and recognize areas that require automation or system enhancement.

In this experiment, students will visit their selected **information system (software or IT firm)**, interact with key users or system personnel, and gather essential details about system objectives, boundaries, inputs, processes, and outputs. The focus is to document how the current system operates and to assess whether a new or improved system is necessary.

This activity aligns with **CO1 (Analyze information systems by applying problem-solving techniques)** and **PO2 (Cognitive: Analyzing, Applying | Affective: Organize)**, as students critically examine real-world systems and propose structured insights through analysis.

2. Requirements

Hardware/Software:

- Computer or laptop with internet connectivity
- Word processor or documentation tool (MS Word / Google Docs / LaTeX)
- Data collection tools (Notebook, Recorder, or Online Survey form)

Other Resources:

- Access permission to the selected organization or system
- Communication with system users, administrators, or developers

3. Procedure

Step 1: Selection of the Target System

- Choose a software-based or IT-enabled system (e.g., Hospital Management System, University Admission Portal, Library Management System, or Payroll Management System).
- Define the system's domain, purpose, and users.

Step 2: Conduct Preliminary Study

- Visit or study the existing information system in operation.
- Identify system **components** such as input, process, output, storage, and feedback mechanisms.
- Observe workflow and user interactions.

Step 3: Gather System Information

- Conduct informal interviews with users and administrators.
- Collect documents such as data flow diagrams (if available), forms, and reports used in the system.
- Note down system characteristics — whether it's manual, computerized, partially automated, or fully digital.

Step 4: Perform Initial Investigation

- Analyze how well the current system meets its objectives.
- Identify **problems** such as data redundancy, lack of integration, processing delays, or user dissatisfaction.
- Determine **system constraints** — technical, operational, and financial.
- Propose **feasibility aspects** (technical, operational, and economic) for possible improvements.

Step 5: Prepare and Submit the Report

- Document findings in a structured report including:
    1. System Overview
    2. Problem Identification
    3. Existing Workflow (with diagrams if possible)
    4. Preliminary Analysis and Feasibility
    5. Suggested Improvements or Next Steps

4. Precautions

- Obtain proper authorization before accessing organizational data.
- Ensure all collected information is used strictly for academic purposes.
- Maintain confidentiality and avoid disclosing sensitive business details.
- Validate findings by cross-checking with multiple users or sources.

5. Conclusion / Discussion

This lab allows students to experience the **first analytical phase of system development**, where understanding an existing system is crucial before proposing improvements or new designs.

By completing this experiment, students will be able to:

- **Analyze** real-world information systems using systematic problem-solving techniques.
- **Identify and document** operational challenges and limitations.
- **Organize and interpret** collected data into meaningful insights for system planning.

This initial investigation forms the basis for further **system requirement analysis, design, and development**, ensuring that future improvements align with actual user needs and organizational goals.

Lab 3

Experiment [CO1:PO2]

Name of the Lab Experiment:

Detailed Analysis of the Information System
(i) Information Gathering for Requirement Analysis
(ii) Analysis of Existing Processes)

1. Introduction / Description

A detailed system analysis is a crucial phase in the **System Development Life Cycle (SDLC)**. It ensures that the analyst fully understands the current system and its requirements before moving on to system design. This lab emphasizes two major tasks: **information gathering** and **process analysis**.

Students will use various **information gathering tools** such as **interviews, questionnaires, and on-site observation** to collect detailed information about the selected system. The gathered data will then be analyzed using structured analysis tools — **Data Flow Diagrams (DFD), Data Dictionary, Decision Trees, and Decision Tables** — to document how the system operates and where improvements are needed.

This experiment helps students **analyze existing information systems by applying systematic problem-solving techniques** in alignment with:

- **CO1:** Analyze information systems by applying problem-solving techniques.
- **PO2:** Cognitive (Analyzing, Applying) & Affective (Organize).

2. Requirements

Hardware/Software:

- Computer or laptop with internet access
- Diagram drawing tool (e.g., Draw.io, Lucidchart, MS Visio, or StarUML)
- Text editor or documentation software (MS Word / Google Docs)

Other Resources:

- Access to the selected organization or software system
- Pre-designed questionnaire and interview templates
- Notebook for recording observations

3. Procedure

Step 1: Selection of Target Information System

- Select the same organization or system chosen in Lab 2 (e.g., Hospital Management System, Library Management System, or Online Admission Portal).
- Clearly define the **scope** of the system (what it does, who uses it, what data it handles).

Step 2: Information Gathering

Use multiple techniques to collect information about the system requirements and processes:

- **Interview:**
  Conduct structured or semi-structured interviews with end-users, system administrators, or stakeholders to understand the system's operations and issues.
- **Questionnaire:**
  Prepare a questionnaire to gather feedback from multiple users about system efficiency, usability, and limitations.
- **On-Site Observation:**
  Observe users performing their daily tasks with the system to understand workflow, inputs, outputs, and challenges faced.

Summarize the collected information in a **requirement summary table** that lists:

- Functional requirements (what the system should do)
- Non-functional requirements (performance, security, usability)
- User and system interactions

Step 3: Structured Analysis of the Existing System

Based on the gathered information, perform a structured analysis using the following tools:

- **Data Flow Diagram (DFD):**
  Create **Level 0 (Context Diagram)** and **Level 1 DFD** to visualize data movement between processes, data stores, and entities.
- **Data Dictionary:**
  Prepare a list of key data elements, their types, and purposes (e.g., Student_ID, Name, Course_Code, etc.).
- **Decision Tree:**
  Represent conditional decisions and their possible outcomes in a tree format to simplify logical flow.
- **Decision Table:**
  Tabulate all possible conditions and corresponding actions for critical decision-making processes.

Step 4: Documentation and Reporting

- Combine all findings into a detailed report including:
    1. System Overview
    2. Methods Used for Information Gathering
    3. Requirement Summary
    4. Structured Analysis (DFD, Data Dictionary, Decision Tree/Table)
    5. Identified System Weaknesses and Improvement Opportunities

4. Precautions

- Ensure that collected data is accurate, verified, and consistent.
- Protect confidentiality of any sensitive or private information.
- Keep records of all interviews and questionnaires for future reference.
- Maintain objectivity while analyzing processes—avoid making assumptions without evidence.

5. Conclusion / Discussion

This experiment develops the student's ability to conduct **comprehensive system analysis** through proper information gathering and structured documentation.

By completing this lab, students will be able to:

- Apply **systematic methods** to collect and analyze information from real systems.
- Use **structured analysis tools** like DFDs and Decision Tables to represent system processes clearly.
- Identify **functional and non-functional requirements** critical for system redesign.
- Strengthen analytical and organizational skills essential for professional system analysts.

This detailed analysis lays the groundwork for the next stages of system design, ensuring that the proposed system improvements are based on a complete understanding of user needs and current process limitations.

Lab 4

Experiment [CO2:PO3]

Name of the Lab Experiment:

Detailed Analysis of the Information System in Terms of
(i) Feasibility Analysis (Economic, Technical, and Behavioral Feasibility)
(ii) Cost-Benefit Analysis

1. Introduction / Description

Before developing or upgrading an information system, it is essential to determine whether the proposed solution is **feasible** in terms of cost, technology, and user adaptability. This experiment focuses on conducting a **feasibility analysis** and performing a **cost-benefit analysis** for the selected information system.

Feasibility analysis assesses whether the proposed system can be successfully developed and implemented within given constraints. It is generally divided into three major categories:

- **Economic Feasibility:** Evaluates whether the expected benefits justify the costs of development and operation.
- **Technical Feasibility:** Assesses whether the existing technology, resources, and expertise can support the system.
- **Behavioral (Operational) Feasibility:** Determines how well the system will be accepted by users and integrated into the organization.

Additionally, a **Cost-Benefit Analysis (CBA)** quantifies the financial viability of the project by comparing the total expected costs with the anticipated benefits. This lab also includes designing a **Weighted Candidate Evaluation Matrix** to rank alternative solutions objectively and prepare a formal **Feasibility Report**.

2. Requirements

Hardware/Software:

- Computer with MS Excel or Google Sheets (for evaluation matrix and calculations)
- Text editor (MS Word / Google Docs) for feasibility and CBA report preparation
- Internet access for data collection and cost estimation

Other Resources:

- Selected organization's or system's background information (from previous labs)
- Financial and operational data (real or hypothetical)

- Evaluation criteria and stakeholder feedback

3. Procedure

Step 1: Review of the Proposed System

- Revisit the system selected in earlier labs (e.g., Library Management System, Online Admission System, or Hospital Management System).
- Clearly define the proposed improvements or new features to be analyzed.

Step 2: Conduct Feasibility Analysis

Perform feasibility study through the following stages:

a) Economic Feasibility

- Identify **development costs** (hardware, software, manpower, training, maintenance).
- Identify **operational benefits** (reduced labor, time savings, better decision-making).
- Calculate **Net Benefit** or **Benefit-Cost Ratio (BCR)**.

Formula 1: Benefit–Cost Ratio (BCR)

$$BCR = \frac{\text{Total Benefits}}{\text{Total Costs}}$$

Interpretation:

- If **BCR > 1**, the project is **economically viable** (benefits exceed costs).
- If **BCR = 1**, the project **breaks even**.
- If **BCR < 1**, the project is **not economically viable** (costs exceed benefits).

b) Technical Feasibility

- Evaluate whether the current infrastructure supports the new system.
- Assess requirements like system compatibility, programming language, database, and technical expertise.

c) Behavioral (Operational) Feasibility

- Evaluate the willingness of users to adopt the system.
- Consider factors like training requirements, user acceptance, and organizational culture.

Step 3: Design Weighted Candidate Evaluation Matrix

Prepare a **Weighted Evaluation Matrix** to objectively compare system alternatives based on specific feasibility factors.

Criteria	**Weight (%)**	**Option A (Score)**	**Option B (Score)**	**Weighted Score (A)**	**Weighted Score (B)**
Economic Feasibility	30	8	7	2.4	2.1
Technical Feasibility	40	9	6	3.6	2.4
Behavioral Feasibility	30	7	8	2.1	2.4
Total	**100**			**8.1**	**6.9**

- The option with the **higher total weighted score** is selected as the most feasible solution.

Step 4: Perform Cost-Benefit Analysis (CBA)

Follow these steps for CBA:

1. **Identify Costs:** Include development, operational, and maintenance costs.
2. **Identify Benefits:** Include tangible benefits (e.g., increased revenue, cost savings) and intangible benefits (e.g., customer satisfaction, improved efficiency).
3. **Classify Benefits:**
   - **Tangible:** Measurable in monetary terms (e.g., reduced labor cost).
   - **Intangible:** Qualitative (e.g., better decision-making).
4. **Compute Net Present Value (NPV) or Payback Period:**
   - **Net Present Value (NPV)**
     [
     $$NPV = \sum \frac{B_t - C_t}{(1 + r)^t}$$
     ]
     where (B_t) = benefits in year *t*, (C_t) = costs in year *t*, and (r) = discount rate.
   - **Payback Period:** Time required for cumulative benefits to equal cumulative costs.
5. Prepare a **Cost-Benefit Table** summarizing annual costs, benefits, and computed results.

Step 5: Prepare Feasibility and CBA Report

Document the following sections in the final report:

1. Introduction and Objective
2. Description of the Proposed System

3. Feasibility Study (Economic, Technical, Behavioral)
4. Weighted Evaluation Matrix
5. Cost-Benefit Analysis (with computations and graphs if applicable)
6. Conclusion and Recommendations

4. Precautions

- Ensure all financial data is accurate and realistic.
- Apply consistent weights and scoring criteria for all options.
- Validate user and technical data through interviews or observations.
- Avoid subjective bias when rating or ranking alternatives.

5. Conclusion / Discussion

This experiment helps students develop the ability to **evaluate the viability of a proposed information system** before implementation.

Through feasibility and cost-benefit analyses, students learn to:

- Assess multiple dimensions of feasibility systematically.
- Design and apply a **Weighted Candidate Evaluation Matrix** for decision-making.
- Perform **quantitative cost-benefit analysis** using suitable financial models.
- Develop professional **feasibility and CBA reports** to support informed system design decisions.

This analytical and design-oriented activity directly supports **CO2 (Design and develop an information system)** and enhances students' understanding of **PO3 (Cognitive – Designing)**.

Lab 5

Experiment [CO3:PO9]

Name of the Lab Experiment:

Design of Alternative Solutions for the Identified Problems

1. Introduction / Description

After completing the phases of **system planning**, **feasibility study**, and **system analysis**, the next step in information system development is to **design alternative solutions** to the identified problems. Designing alternatives allows system analysts to evaluate different ways of achieving the same objectives and choose the most effective, efficient, and user-friendly option.

In this experiment, students will work individually and collaboratively in teams to design **candidate system solutions** based on the earlier analysis. Each solution will be supported with **structured design tools** such as:

- **Data Flow Diagrams (DFDs)** for functional representation,
- **Entity Relationship Diagrams (ERDs)** for data modeling,
- **System architecture charts or flowcharts** for process overview.

This lab emphasizes teamwork, creativity, and design thinking to address **real-life complex information system problems**, ensuring that the proposed system is both technically viable and operationally effective.

2. Requirements

Hardware/Software:

- Computer with standard configuration
- Diagram design tools: **Draw.io**, **Lucidchart**, **MS Visio**, or **StarUML**
- Text editor (MS Word / Google Docs) for documentation
- Collaboration platform: **GitHub**, **Google Drive**, or **Microsoft Teams**

Other Resources:

- Problem statements and system analysis reports from previous labs
- Team collaboration plan and assigned roles
- Templates for DFDs and ERDs

3. Procedure

Step 1: Review the Identified Problems

- Revisit the system identified in earlier labs (e.g., library management system, student registration system, hospital management system).
- Summarize the key **issues, limitations, and user requirements** discovered during feasibility and analysis stages.
- Discuss with your team possible design directions that address these challenges.

Step 2: Identify and Define Candidate Solutions

- Propose **at least two to three alternative system designs** that could solve the identified problems.
- For each design alternative, clearly describe:
    - **Objective and scope** of the design
    - **Required technologies and resources**
    - **User interaction or interface concept**
    - **Expected advantages and trade-offs**

Prepare a summary table comparing these alternatives.

Criteria	Alternative 1	Alternative 2	Alternative 3
Technology Used	Web-based	Mobile App	Hybrid
Cost	Medium	Low	High
User Accessibility	High	Medium	High
Implementation Time	Moderate	Short	Long
Feasibility	✓	✓✓	✓

Step 3: Develop Structured Design Models

A. Data Flow Diagram (DFD):

- Draw **Level 0 (Context Diagram)** to represent the system's interaction with external entities.
- Draw **Level 1 DFD** to decompose the system into major processes.
- Identify data stores, data flows, and processes for each alternative design.

B. Entity Relationship Diagram (ERD):

- Identify system entities, attributes, and relationships.
- Design an **ER model** showing one-to-many and many-to-many relationships.
- Normalize the data up to 3NF for efficiency and clarity.

C. Other Representations (if applicable):

- **System Flowchart:** to depict data and control flow among modules.

- **Use Case Diagram:** to describe interactions between users (actors) and system functions.
- **Architecture Diagram:** to show overall system structure (client-server, 3-tier, or cloud-based).

Step 4: Team Collaboration and Discussion

- Assign roles within the team (e.g., system designer, data modeler, documentation lead).
- Collaboratively discuss and refine each proposed design alternative.
- Record discussions, decisions, and justifications for the chosen approach.
- Ensure that all members contribute to design modeling and documentation.

Step 5: Documentation and Reporting

Prepare a **Design Report** that includes:

1. Introduction and Objectives
2. Identified Problems (Summary from previous labs)
3. Description of Candidate Solutions
4. DFDs and ERDs for Each Design
5. Comparison of Alternative Designs
6. Team Discussion and Chosen Solution
7. Conclusion and Recommendations

Include all diagrams and relevant screenshots in the report.

4. Precautions

- Maintain consistency in notations and naming conventions across all diagrams.
- Validate DFDs and ERDs for logical accuracy and completeness.
- Avoid redundant entities or processes.
- Use collaborative version control tools (e.g., GitHub) to manage team contributions.
- Respect teamwork ethics — ensure all members actively participate.

5. Conclusion / Discussion

This experiment demonstrates how to **design and evaluate multiple alternative solutions** for complex information system problems.

By completing this lab, students will:

- Apply **design methodologies** to propose feasible system solutions.
- Use structured modeling tools like **DFD** and **ERD** to represent system structure and data flow.
- Strengthen their **problem-solving and decision-making** skills.

- Work effectively both **individually and collaboratively** to handle multidisciplinary design challenges.

This lab directly supports **CO3 (Solve real-life complex problems individually and collaboratively)** and **PO9 (Teamwork and leadership skills)** by promoting creativity, critical analysis, and cooperative design practices in real-world contexts.

Lab 6

Experiment [CO3:PO9]

Name of the Lab Experiment:

Finalization of the Design for the Targeted Information System

1. Introduction / Description

After the analysis and evaluation of alternative system solutions, the next crucial step in information system development is to **finalize the design** of the targeted system. This stage transforms the logical concepts into a detailed and implementable system design specification.

In this experiment, students will perform both **logical** and **physical design** activities of the targeted information system. They will apply structured design methodologies to represent system hierarchy, data flow, and inter-module relationships through tools like **functional decomposition**, **structure charts**, and discussions on **module coupling** and **cohesion**.

Additionally, students will explore **major development activities**, including **database design**, **program design**, **system and program testing preparation**, and **interface specifications**. This experiment aims to simulate the collaborative process of transforming analytical results into a well-defined, ready-to-implement system design.

2. Requirements

Hardware/Software:

- Standard computer system
- Design tools: **Draw.io**, **Lucidchart**, **StarUML**, **MS Visio**
- Database tool: **MySQL Workbench**, **ERDPlus**, or **Oracle Data Modeler**
- Code design or documentation tool: **MS Word**, **Google Docs**, or **Visual Paradigm**
- Collaboration tools: **GitHub**, **Google Drive**, or **Microsoft Teams**

Other Resources:

- Finalized feasibility report and system analysis results from previous labs
- Team collaboration plan
- Access to sample datasets (if applicable)

3. Procedure

Step 1: Review the Approved System Design

- Review the results from the previous experiments (alternative solutions).
- Identify and confirm the **targeted system** to be finalized for design.
- Summarize the functional and non-functional requirements for the chosen system.

Step 2: Conduct Logical Design

- Develop the **logical model** of the system representing *what* the system will do without specifying *how* it will be implemented.
- Include the following components:
    - **Input and Output Design:** Identify system inputs and expected outputs.
    - **Process Specifications:** Define key processes, their objectives, and dependencies.
    - **Data Design:** Define data elements, data structures, and relationships.
    - **Functional Decomposition Diagram (FDD):** Break the system into major functions and sub-functions.

Step 3: Perform Physical Design

- Convert logical designs into physical representations that specify *how* the system will operate.
- Include:
    - **Database Design:**
        - Define entity tables, primary and foreign keys, and normalization (up to 3NF).
        - Prepare **ERD** and **Relational Schema** diagrams.
    - **System Architecture:**
        - Specify system components (client/server, web-based, etc.) and their communication flow.
    - **Program Design:**
        - Outline modular programs or procedures to implement the identified processes.
        - Specify program input/output files, control logic, and interfaces.

Step 4: Design Structured Charts

- Draw the **Structure Chart** showing the hierarchy and relationships among different modules of the system.
- Indicate control and data flow among modules.
- Ensure modular independence by maintaining **high cohesion** within modules and **low coupling** between modules.

Discuss:

- **Module Coupling:** Degree of interdependence between modules (aim for minimal coupling).
- **Module Cohesion:** Degree to which elements of a module belong together (aim for strong cohesion).

Step 5: Discuss Major Development Activities

Conduct and document discussions on the following **development activities** involved in implementing the finalized system design:

1. **Database Design:** Logical and physical database structure, normalization, and indexing strategies.
2. **Program Design:** Modular breakdown of application logic, algorithms, and pseudocode.
3. **System and Program Testing Preparation:**
    - Identify test cases, input data, expected outputs, and error handling procedures.
4. **System Interface Specification:**
    - Design internal and external interfaces for data exchange and user interaction.
    - Specify screen layouts, menu structures, and API integrations (if any).

Step 6: Documentation and Reporting

Prepare a comprehensive **System Design Report** that includes:

1. Title of Experiment and Objective
2. Description of Targeted System
3. Logical and Physical Design Details
4. Functional Decomposition Diagram
5. Structure Chart with Explanation
6. Discussion on Module Coupling and Cohesion
7. Database Design (ERD, Relational Schema)
8. Summary of Major Development Activities
9. Conclusion

Attach all diagrams and related design documentation in the final report.

4. Precautions

- Maintain consistency between logical and physical design elements.
- Validate all diagrams for accuracy, completeness, and clarity.
- Ensure modularity, low coupling, and high cohesion in system structure.
- Verify that database design meets normalization standards.
- Encourage active team participation during design discussions and documentation.

5. Conclusion / Discussion

This experiment consolidates all prior analysis and design phases into a finalized, detailed system design ready for implementation.

By completing this lab, students will:

- Apply structured design methodologies to finalize system specifications.
- Develop skills in **logical and physical design**, **functional decomposition**, and **modular structure creation**.
- Gain hands-on experience with **database and program design** activities.
- Demonstrate teamwork and leadership through collaborative system design tasks.

This lab reinforces **CO3 (Solving real-life complex problems individually and collaboratively)** and **PO9 (Individual work and teamwork)** by integrating analytical, design, and communication skills in the development of a complete information system.

Lab 7

Exam

Quiz, Presentation/ Viva voce will be held during the Lab time.

1. Introduction / Description

This lab session serves as the **assessment phase** of the Information System Analysis and Design (ISAD) Laboratory course.

Students will participate in a **Quiz**, **Presentation**, and **Viva Voce**, which will evaluate both their theoretical understanding and practical application of information system concepts covered throughout the semester.

The session aims to assess students' ability to:

- Apply problem-solving and analytical skills to real-world system scenarios.
- Design and communicate effective system solutions.
- Demonstrate teamwork, leadership, and professional communication.

This evaluation provides a comprehensive understanding of each student's learning outcomes and readiness to apply ISAD principles in real-life software development and analysis environments.

2. Requirements

- Prepared lab notebook and all completed experiment reports.
- Final system analysis and design report (individual or group).
- Presentation slides summarizing the project or experiment outcomes.
- Stable internet connection and access to GitHub/Drive (if presenting system design files).

3. Procedure

Part A: Quiz

- The quiz will include **short questions, multiple-choice questions, and analytical scenarios** covering all major ISAD topics:
    - System development life cycle (SDLC)
    - Information gathering tools
    - Feasibility analysis and cost-benefit analysis
    - DFD, ERD, structure chart, and system design
- The total marks will be **20**, and the duration will be **20–25 minutes**.
- The quiz assesses students' **conceptual clarity and analytical ability** (aligned with CO1 & PO2).

Part B: Presentation

- Each individual or group will present their **final system design project or analysis report**.
- Presentation duration: **5–10 minutes per team**.
- The presentation should include:
    - Problem identification and objectives
    - System analysis and feasibility summary
    - Proposed design models (DFD, ERD, structure charts, etc.)
    - Implementation or testing overview (if applicable)
    - Challenges faced and solutions applied
- Students should use **PowerPoint slides or design tools** for visualization.
- This section emphasizes **CO2 and PO3**, focusing on design thinking and communication skills.

Part C: Viva Voce

- The viva voce will be conducted individually.
- Questions will be asked on:
    - Concepts and methods used in experiments
    - Tools and techniques applied (e.g., GitHub, Draw.io, SQL tools)
    - Interpretation of results and analysis logic
- The viva assesses **individual understanding, clarity of concept, and the ability to apply knowledge practically**.
- It directly supports **CO3 and PO9**, highlighting teamwork and individual responsibility.

4. Precautions

- Revise all completed experiments thoroughly before the session.
- Ensure all reports and project files are up to date and properly organized.
- Maintain punctuality and professionalism during the quiz and presentation.
- Collaborate respectfully with teammates and follow instructor guidance.

5. Conclusion / Discussion

This lab integrates assessment activities that holistically evaluate students' learning outcomes across all ISAD course objectives.

Through the **quiz, presentation, and viva voce**, students demonstrate:

- Conceptual understanding and analytical ability.
- Design proficiency in developing information systems.
- Communication, leadership, and teamwork skills essential for professional practice.

By successfully completing this lab, students will showcase their preparedness to apply **Information System Analysis and Design** methodologies to real-world system development challenges.